

Practice Questions for Pivot Rules

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1 Questions

1. In the Simplex method, what is the significance of a negative coefficient in the objective function row after a pivot operation, and how does it relate to the optimality of the current solution?
 - a. A negative coefficient indicates the solution is optimal; further iterations are unnecessary.
 - b. A negative coefficient indicates the solution is not optimal; the corresponding non-basic variable should be increased.
 - c. A negative coefficient has no significance; it's simply a byproduct of the pivot operation.
 - d. A negative coefficient indicates an infeasible solution; the problem needs to be reformulated.
 - e. A negative coefficient indicates degeneracy; the simplex method may cycle.
2. Explain the concept of a "pivot" in the simplex method and its geometric interpretation in relation to the feasible region.
 - a. A pivot is a simple arithmetic operation; it has no geometric significance.
 - b. A pivot involves selecting a new basic variable and solving for it; geometrically, it represents moving from one vertex to an adjacent vertex in the feasible region.
 - c. A pivot is the process of finding the optimal solution; geometrically, it's finding the highest point in the feasible region.
 - d. A pivot is a random selection of variables; geometrically, it's a random jump within the feasible region.
 - e. A pivot is the initial step of the simplex method; geometrically, it's selecting the origin as the starting point.
3. Describe the role of the maximum coefficient rule in selecting the entering variable in the simplex method, and discuss its limitations.
 - a. The maximum coefficient rule always selects the optimal entering variable; it has no limitations.
 - b. The maximum coefficient rule selects the variable with the largest coefficient in the objective function; it's efficient but may not always lead to the fewest iterations.
 - c. The maximum coefficient rule selects a random variable; it's simple but inefficient.
 - d. The maximum coefficient rule selects the variable with the smallest coefficient; it's cautious but may be slow to converge.
 - e. The maximum coefficient rule is only used in the initial iteration; it's replaced by other rules in subsequent iterations.

4. What is the significance of the minimum ratio test in the simplex method, and how does it ensure feasibility?
 - a. The minimum ratio test is used to select the entering variable; it ensures optimality.
 - b. The minimum ratio test is used to select the leaving variable; it ensures feasibility.
 - c. The minimum ratio test is an optional step; it doesn't affect feasibility or optimality.
 - d. The minimum ratio test is used to check for degeneracy; it ensures the algorithm doesn't cycle.
 - e. The minimum ratio test is used to check for unboundedness; it ensures the problem has a finite solution.
5. In the Simplex Method, using the maximum coefficient rule, which variable should enter the basis first in the linear program presented on Slide 1, given the objective function $\zeta = 3x_1 + 2x_2$?
 - a. x_2
 - b. x_1
 - c. w_1
 - d. w_2
 - e. w_3
6. On Slide 8, while applying the simplex method and increasing x_1 while keeping $x_2 = 0$, which constraint determines the maximum feasible value of x_1 ?
 - a. $w_1 \geq 0$
 - b. $w_2 \geq 0$
 - c. $w_3 \geq 0$
 - d. $x_1 \geq 0$
 - e. $x_2 \geq 0$
7. On Slide 11, what is the maximum value of x_2 allowed by the constraints, given that $w_3 = 0$?
 - a. 6.8
 - b. 8
 - c. 12
 - d. 2
 - e. 10
8. In the Simplex Method, after the pivot operation on Slide 12, which variable has left the basis?
 - a. x_1
 - b. x_2
 - c. w_1
 - d. w_2

- e. w_3
9. a. After introducing slack variables w_1, w_2, w_3 , what is the equation for w_1 ?
- A. $w_1 = 12 - x_1 + 3x_2$
 - B. $w_1 = 12 + x_1 - 3x_2$
 - C. $w_1 = 12 - x_1 - 3x_2$
 - D. $w_1 = -12 + x_1 - 3x_2$
 - E. $w_1 = -12 - x_1 + 3x_2$
- b. In the initial basic feasible solution (IBFS) obtained by setting the non-basic variables (x_1, x_2) to zero, what is the value of w_2 ?
- A. 0
 - B. 8
 - C. 12
 - D. 10
 - E. 5
- c. What is the value of the objective function ζ at the IBFS?
- A. 12
 - B. 22
 - C. 10
 - D. 8
 - E. 0
- d. Which variable would enter the basis in the first iteration of the simplex method (using the maximum coefficient rule)?
- A. w_1
 - B. w_2
 - C. w_3
 - D. x_1
 - E. x_2
- e. On Slide 8, increasing x_1 while keeping $x_2 = 0$, which constraint determines the maximum feasible value of x_1 ?
- A. $-x_1 + 3x_2 \leq 12$
 - B. $x_1 + x_2 \leq 8$
 - C. $2x_1 - x_2 \leq 10$
 - D. $x_1 \geq 0$
 - E. $x_2 \geq 0$
10. In the simplex method, using the maximum coefficient rule, what is the next variable to enter the basis after the initial step where $x_1 = x_2 = 0$ in the linear program presented on Slide 1, given the objective function $\zeta = 3x_1 + 2x_2$?

- a. w_1
- b. x_1
- c. x_2
- d. w_2
- e. w_3

2 Answers

1. In the Simplex method, what is the significance of a negative coefficient in the objective function row after a pivot operation, and how does it relate to the optimality of the current solution?
 - a. A negative coefficient in the objective function row after a pivot means that increasing the corresponding non-basic variable would decrease the objective function value. Since all variables are non-negative, this implies that the current solution is optimal.
 - b. While a negative coefficient indicates the solution is not optimal, it's the *positive* coefficients that indicate the possibility of improvement by increasing the corresponding non-basic variable.
 - c. Negative coefficients are crucial in determining optimality. They indicate that the current solution is optimal because increasing the corresponding non-basic variable would decrease the objective function value.
 - d. An infeasible solution would be indicated by negative values in the right-hand side of the constraints, not negative coefficients in the objective function row.
 - e. Degeneracy is related to multiple basic variables having a value of zero, which can lead to cycling. Negative coefficients in the objective function row are related to optimality, not degeneracy.
2. Explain the concept of a "pivot" in the simplex method and its geometric interpretation in relation to the feasible region.
 - a. The pivot operation is not just a simple arithmetic operation; it has a significant geometric interpretation in the context of the feasible region.
 - b. A pivot operation involves selecting an entering variable (non-basic) and a leaving variable (basic) and then performing row operations to express the entering variable in terms of the remaining non-basic variables. Geometrically, this corresponds to moving from one extreme point (vertex) of the feasible region to an adjacent extreme point.
 - c. Finding the optimal solution is the overall goal of the simplex method, but a pivot is a single step in that process. Geometrically, it's not simply finding the highest point, but rather moving along the edges of the feasible region.
 - d. Pivoting is a systematic process, not a random selection. The choice of entering and leaving variables is based on specific rules to ensure improvement in the objective function and maintain feasibility.
 - e. The initial step involves setting non-basic variables to zero, which corresponds to selecting a specific vertex (often the origin if feasible), but the pivot operation comes later in the algorithm.
3. Describe the role of the maximum coefficient rule in selecting the entering variable in the simplex method, and discuss its limitations.
 - a. The maximum coefficient rule is a heuristic; it doesn't guarantee finding the optimal entering variable in the fewest number of iterations.
 - b. The maximum coefficient rule prioritizes the variable with the largest positive coefficient in the objective function row, aiming for the greatest immediate improvement in the objective function value. However, this greedy approach doesn't guarantee the fewest iterations to reach the optimal solution.
 - c. The maximum coefficient rule is not random; it's a deterministic rule based on the objective function coefficients.

- d. The maximum coefficient rule uses the largest positive coefficient, not the smallest.
 - e. The maximum coefficient rule can be used in any iteration, although other pivot rules might be more efficient in some cases.
4. What is the significance of the minimum ratio test in the simplex method, and how does it ensure feasibility?
- a. The entering variable is selected based on the objective function (e.g., maximum coefficient rule), not the minimum ratio test.
 - b. The minimum ratio test is used to determine the leaving variable from the basis. It ensures that after the pivot operation, all variables remain non-negative, thus maintaining feasibility.
 - c. The minimum ratio test is a crucial step in the simplex method to guarantee feasibility.
 - d. The minimum ratio test doesn't directly address degeneracy, although degeneracy can affect the choice of the leaving variable.
 - e. Unboundedness is detected when there is no positive ratio in the minimum ratio test, indicating that a variable can be increased indefinitely without violating constraints.
5. In the Simplex Method, using the maximum coefficient rule, which variable should enter the basis first in the linear program presented on Slide 1, given the objective function $\zeta = 3x_1 + 2x_2$?
- a. While x_2 is a variable in the problem, its coefficient (2) is smaller than that of x_1 (3). The maximum coefficient rule prioritizes the variable with the largest coefficient.
 - b. The maximum coefficient rule dictates that the variable with the largest coefficient in the objective function should enter the basis first. In this case, x_1 has a coefficient of 3, which is larger than the coefficient of x_2 (2). Therefore, x_1 enters the basis first.
 - c. Slack variables (w_1, w_2, w_3) are initially basic variables. The maximum coefficient rule applies to non-basic variables.
 - d. Similar to w_1 , w_2 is a slack variable and initially basic. The maximum coefficient rule selects from non-basic variables.
 - e. Similar to w_1 and w_2 , w_3 is a slack variable and initially basic. The maximum coefficient rule selects from non-basic variables.
6. On Slide 8, while applying the simplex method and increasing x_1 while keeping $x_2 = 0$, which constraint determines the maximum feasible value of x_1 ?
- a. The constraint $w_1 \geq 0$ gives $12 + x_1 \geq 0$, which implies $x_1 \geq -12$. This is not a limiting constraint.
 - b. The constraint $w_2 \geq 0$ gives $8 - x_1 \geq 0$, which implies $x_1 \leq 8$. This is less restrictive than $w_3 \geq 0$.
 - c. The constraints are $w_1 = 12 + x_1 - 3x_2$, $w_2 = 8 - x_1 - x_2$, and $w_3 = 10 - 2x_1 + x_2$. Since $x_2 = 0$, we have $w_2 = 8 - x_1$ and $w_3 = 10 - 2x_1$. For feasibility, $w_2 \geq 0 \Rightarrow x_1 \leq 8$ and $w_3 \geq 0 \Rightarrow x_1 \leq 5$. The most restrictive constraint is $w_3 \geq 0$, limiting x_1 to a maximum of 5.
 - d. The constraint $x_1 \geq 0$ is a non-negativity constraint and does not determine the maximum feasible value of x_1 in this context.
 - e. The constraint $x_2 \geq 0$ is given as a condition ($x_2 = 0$) and does not determine the maximum feasible value of x_1 .

7. On Slide 11, what is the maximum value of x_2 allowed by the constraints, given that $w_3 = 0$?
- This value comes from the constraint $w_1 \geq 0$, which is less restrictive than $w_2 \geq 0$.
 - This value is unrelated to the constraints given on Slide 11.
 - This value is unrelated to the constraints given on Slide 11.
 - With $w_3 = 0$, the constraints become: $x_1 = 5 + \frac{1}{2}x_2$, $w_1 = 17 - \frac{5}{2}x_2$, and $w_2 = 3 - \frac{3}{2}x_2$. For feasibility, $w_1 \geq 0 \Rightarrow x_2 \leq \frac{34}{5} = 6.8$ and $w_2 \geq 0 \Rightarrow x_2 \leq 2$. The most restrictive constraint is $w_2 \geq 0$, limiting x_2 to a maximum of 2.
 - This value is unrelated to the constraints given on Slide 11.
8. In the Simplex Method, after the pivot operation on Slide 12, which variable has left the basis?
- x_1 remains in the basis after this pivot operation.
 - x_2 enters the basis in this iteration.
 - w_1 remains in the basis after this pivot operation.
 - Slide 12 describes the interchange of x_2 and w_2 . x_2 enters the basis, and w_2 leaves the basis.
 - w_3 remains in the basis after this pivot operation.
9. a. After introducing slack variables w_1, w_2, w_3 , what is the equation for w_1 ?
- This equation incorrectly uses the wrong signs for x_1 and $3x_2$.
 - The slack variable w_1 is added to the first constraint to transform the inequality into an equality: $-x_1 + 3x_2 + w_1 = 12$. Solving for w_1 , we get $w_1 = 12 + x_1 - 3x_2$.
 - This equation incorrectly uses the wrong sign for $3x_2$.
 - This equation incorrectly uses the wrong signs for the constant and the variables.
 - This equation incorrectly uses the wrong signs for the constant and x_1 .
- b. In the initial basic feasible solution (IBFS) obtained by setting the non-basic variables (x_1, x_2) to zero, what is the value of w_2 ?
- This is incorrect; the IBFS is obtained by setting the non-basic variables to zero, but w_2 is a basic variable.
 - Setting $x_1 = 0$ and $x_2 = 0$ in the equation $w_2 = 8 - x_1 - x_2$ gives $w_2 = 8 - 0 - 0 = 8$.
 - This is the value of w_1 in the IBFS, not w_2 .
 - This is the value of w_3 in the IBFS, not w_2 .
 - This is not the value of any of the slack variables in the IBFS.
- c. What is the value of the objective function ζ at the IBFS?
- This is incorrect; the objective function value at the IBFS is 0.
 - This is the optimal value of the objective function, not the initial value.
 - This is the value of w_3 at the IBFS.
 - This is the value of w_2 at the IBFS.
 - At the IBFS, $x_1 = x_2 = 0$, so $\zeta = 3(0) + 2(0) = 0$.

- d. Which variable would enter the basis in the first iteration of the simplex method (using the maximum coefficient rule)?
- A. Slack variables are typically basic variables initially.
 - B. Slack variables are typically basic variables initially.
 - C. Slack variables are typically basic variables initially.
 - D. The maximum coefficient rule selects the variable with the largest coefficient in the objective function. Since the coefficient of x_1 (3) is larger than the coefficient of x_2 (2), x_1 enters the basis.
 - E. While x_2 has a positive coefficient, x_1 's coefficient is larger.
- e. On Slide 8, increasing x_1 while keeping $x_2 = 0$, which constraint determines the maximum feasible value of x_1 ?
- A. This constraint is not the most restrictive when $x_2 = 0$.
 - B. This constraint gives $x_1 \leq 8$, which is less restrictive than $x_1 \leq 5$.
 - C. With $x_2 = 0$, the constraint becomes $2x_1 \leq 10$, which implies $x_1 \leq 5$. This is the most restrictive constraint on x_1 in this case.
 - D. This constraint is always satisfied since x_1 is non-negative.
 - E. This constraint is always satisfied since x_2 is non-negative and is already set to 0.
10. In the simplex method, using the maximum coefficient rule, what is the next variable to enter the basis after the initial step where $x_1 = x_2 = 0$ in the linear program presented on Slide 1, given the objective function $\zeta = 3x_1 + 2x_2$?
- a. w_1 is a slack variable and is already in the basis in the initial step. The maximum coefficient rule selects a variable *not* currently in the basis.
 - b. The maximum coefficient rule dictates that the variable with the largest coefficient in the objective function should enter the basis first. In this case, the objective function is $\zeta = 3x_1 + 2x_2$, and x_1 has the largest coefficient (3). Therefore, x_1 is the next variable to enter the basis.
 - c. While x_2 has a positive coefficient in the objective function, it is smaller than the coefficient of x_1 . The maximum coefficient rule prioritizes the variable with the largest coefficient.
 - d. w_2 is a slack variable and is already in the basis in the initial step. The maximum coefficient rule selects a variable *not* currently in the basis.
 - e. w_3 is a slack variable and is already in the basis in the initial step. The maximum coefficient rule selects a variable *not* currently in the basis.